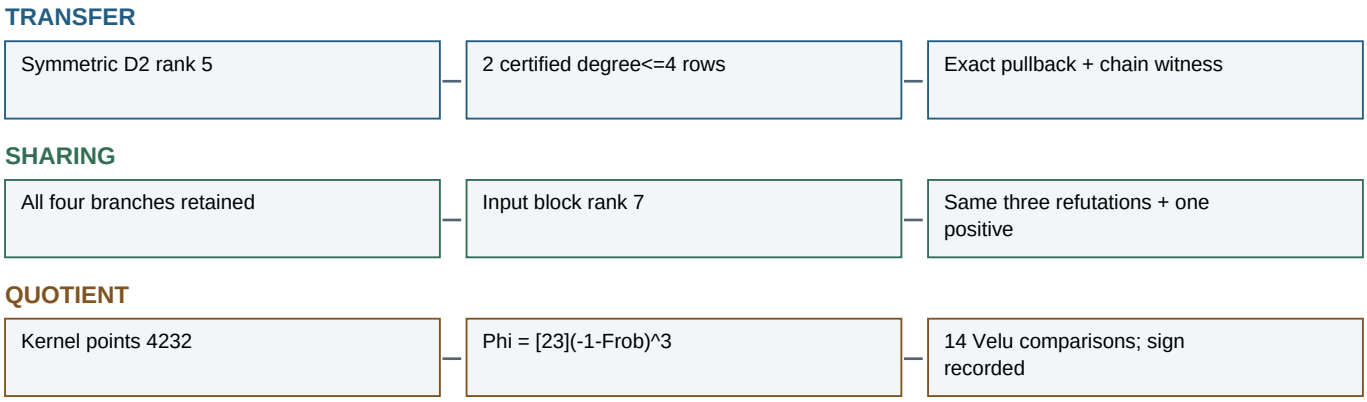
Checked components for three ECDLP index-calculus follow-ons

29 October 2026. **Six native conformance tests, strict Clippy and seven conditional Lean lemmas pass. Complete query/branch-set costs and speedups remain unknown. No cloud experiment was launched.**

This advances the three published proposals to executable, source-linked components. It does not complete their priced pilots, the original frozen programme or a novelty audit. The native crate is standalone: its checked source does not depend on the older review tree's crypto library. Earlier sixteen-test library results keep their old-source scope; they are not a test of the current merged library.



Fixed software controls. Complete costs/speedups unknown. No scaling or crossover claim.

1. Selective symmetric-to-chain transfer

The new component builds the actual b=1 binary symmetric S4 system over the product spaces V, V^2 and V^3, retaining exact Boolean source multipliers for every degree-two closure row. Its substitution into three factor-base legs is checked coefficient-for-coefficient. An exact quadratic-resultant Bezout witness then expresses each pulled-back row in the original two-S3 chain. This avoids assuming a nonzero leading coefficient or dropping exceptional roots.

For quadratics f=az^2+bz+c, g=dz^2+ez+f0, put u=ae-bd and v=af0-cd. The checked identity is

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R = v^2-u(bf0-ce)
  = (duz+eu-dv)f + (av-auz-bu)g.
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It is proved in a commutative ring and reduces correctly to characteristic2. Certificate **IDC1h02c42d4a878d05d0** binds that identity and the checked Lean source. The Boolean representation and its substitution remain separately checked native components, not a full Lean formalization of the solver.

Low-degree selection is the kernel of projection onto terms above degree4. It considers combinations of rows; simply discarding displayed high-degree basis rows missed useful cancellations in the initial development version. The original records were preserved and the amended component adds a cancellation control, a corrupted-certificate control and a genuine prime-subgroup positive control.

All controls use E0: y^2+xy=x^3+1 over F32, modulus z^5+z^2+1. Three summands in V give six x-bits; the chain adds five intermediate-point bits. The last row uses the first nonidentity [4]P in the deterministic point order, V=<1,x([4]P)>=<1,4>, and target 3, whose abscissa is2.

V basis	target x	symmetric D2 rank	exact degree<=4 image rank	maximum pullback degree	verified subgroup x-assignments
1,2	0	30	12	6	0
1,2	1	5	3	6	0
1,4	2	5	2	6	1

For each row, all 64 x-assignments and all 2048 chain assignments are checked. The chain has respectively 9, 11 and 1 solutions; every selected consequence preserves them. The first two controls have ambient-curve relations but no prime-subgroup positive relation. They cannot serve as prime-subgroup query evidence. The added third control supplies that missing positive check.

Conclusion in scope: two degree-at-most-four consequences survive exact selection in the prime control, despite individual pullbacks reaching degree6. This supplies a valid transfer mechanism. It does not show that these rows are new to the direct chain's same-degree closure, reduce solving degree, or lower complete verified-query cost. Translation, certificate replay, failed roots, signed lifting and all verification still need pricing.

2. Exact sharing across all guessed branches

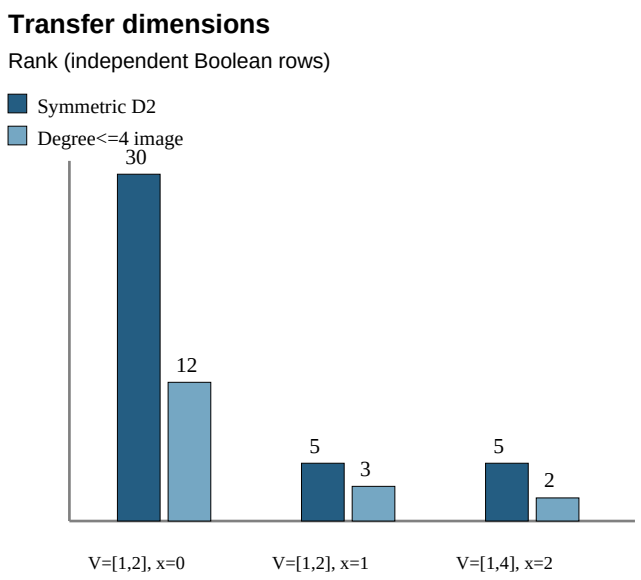
The component fixes the first two x-bits and retains all four leaves. It computes the exact intersection of residual row spaces and replays each common row against the original generators separately in every leaf. An equal-rank counterexample, span{x} versus span{y}, has zero intersection; equal ranks are never used as an equivalence certificate.

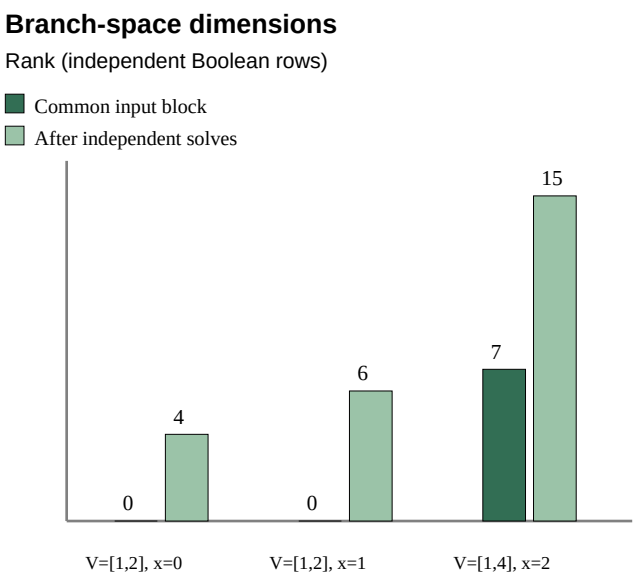
A separate path identifies literally common Macaulay input rows, eliminates their common block once, and then finishes every leaf with its own source certificate. It takes the residual generators as input and does not require the reference leaf solutions. The audit also computes independent closures for comparison; that reference work is not a free preprocessing step.

V basis; target x	independent leaf ranks	intersection after solves	common input rows	common input block rank
1,2;0	12,14,13,12	4	0	0
1,2;1	14,12,14,14	6	0	0
1,4;2	16,16,15,16	15	10	7

In the prime control the branch outcomes are exactly **refuted, refuted, one positive assignment, refuted**. The shared-input and independent paths have equal row spaces and preserve every outcome. No leaf is omitted. The active system has only four Boolean variables and closure degree4; this finite completeness is not evidence of a degree4 law at larger sizes.

Conclusion in scope: a seven-row pre-elimination block exists for the prime control. The two other controls have no literally common input block, even though their post-solve intersections are nonzero. Post-solve rank15 in the prime control includes information already obtained by refuting three leaves; it is not a zero-cost shared solver. Compilation, common-block discovery, per-leaf certificates/replay, memory and the remaining solves may erase any saving. No full branch-set cost improvement is claimed.





Native conformance records: transfer.json / sharing.json; three fixed controls; exact ranks, no uncertainty estimate.

Different spaces/stages are labelled. These dimensions are neither costs nor speedups; post-solve sharing is diagnostic.

3. An explicit quotient for the existing large-cofactor E0 fixture

The existing fixture is binary, not a prime-field curve: degree 22, modulus z^22+z+1 (integer 4194307), E0 with a2=0,b=1, recorded ambient order 4193912=991*4232 and known generator (97814,2630359). That ambient order is an input retained from the earlier audited fixture; this component does not prove a new cardinality result.

Let pi be coordinate Frobenius. E0(F2) consists of infinity,(0,1),(1,0),(1,1), so its F2 trace is -1. The standard Frobenius characteristic equation is pi^2+pi+[2]=0. Hence V=-1-pi is its Verschiebung, with pi V=V pi=[2]. The concrete maps are

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Phi = [23] V^3,      Psi = [23] pi^3,
Psi Phi = [23^2] (pi V)^3 = [4232].
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The general polynomial identity is certificate **IDC1hf913cc1220ef209b**. Its specialization is checked in Lean under the explicit commuting/FV=2 assumptions. The actual elliptic-curve geometry is standard source material and is not formalized by that ring lemma.

Here is the fixed-fixture kernel argument. Conditional on the recorded order and coprimality, the h-multiplication kernel has4232 points. The constructed set contains4232 distinct points killed by[h], so it covers that kernel. The native check kills every one with Phi. Conversely, Psi Phi=[h] implies that every Phi-kernel point is killed by[h]. The two kernels are therefore equal. Phi is nonzero on the known order 991 generator, hence injective on that prime subgroup. The written argument uses standard finite-group and Frobenius facts; the Lean kernel/subgroup lemmas expose their coverage and injectivity hypotheses rather than assuming the desired Phi-kernel.

Using standard isogeny degree facts, V is separable of degree 2 and[23] has degree 529; Phi is an explicit separable degree 4232 quotient on E0. This geometric degree statement is not a full Lean proof. The native map uses three V steps and scalar multiplication by23 and does not enumerate 4232 kernel corrections during evaluation. This is a construction mechanism, not a measured speedup or an implicit algebraic relation oracle.

An independent Vélu formula implementation provides the comparison. Complete small controls over F32,F64,F128 enumerate 44, 56 and 116 rational points, verify all 1936, 3136 and 13456 homomorphism pairs, and have respective kernel sizes 4, 8 and 4 and image sizes 11, 7 and 29. Lifting is also checked against every affine pair. Both map implementations share the native field/point backend; no external independent arithmetic implementation or runner is claimed.

For the degree 22 fixture, all 4232 kernel points pass, while the retained 2116-point incomplete-span control remains incomplete. Fourteen nonkernel and kernel-translate comparisons agree with **minus Phi** after the Vélu target normalization y'=y+1. The normalized target has the same E0 parameters. This is finite comparison evidence, not a proved global equality of expanded rational maps. The complementary identity is also checked on the recorded point controls.

Conclusion in scope: the proposed quotient now has a concrete E0-specific construction and verified component controls. An isogeny preserves the unknown-log problem; it supplies no scalar solver. The image factor base, algebraic map encoding/degree, exceptional cases, signed lifting, duplicate images, independent relation rank and every cold phase still need an oracle implementation and complete comparison. The small-cofactor challenge-family atlas closures are not reopened.

Verification and precise completion status

- Six release library tests pass, no failures/ignores; strict all-target Clippy with -D warnings passes. These are local development checks, with no admitted benchmark timing or calibrated operation comparison.

- Seven Lean 4.34.1 lemmas pass. Five use only propext, Classical.choice and Quot.sound; two conditional kernel/transport bridges use no axioms. No sorryAx, custom research axiom or native_decide. Fresh lean4checker is unavailable. See the proof map for missing representation/geometry proofs.

- Each v1 identity passed 32 issuer-derived plus 64 separately seeded checker points, and an added-one false identity was rejected. The checked Lean source hashes match the proof pointers. This is same-session verification, not external independent replay or leaderboard admission.

- Executable components/certificates/proof checks: verified in the stated finite/conditional scope. Full-query and full-branch-set prices: not measured. Complete quotient relation oracle: not implemented. Growing-size scaling, external-tool comparison and crossover: not attempted here.

Final raw component records, the exact input-file inventory, proof hashes and error/repair logs are in ../../followons_20261029/validation/. Initial missing dependency/serialization and proof-zero-step issues were corrected, not waived. The incomplete basis-row-only selector was replaced by exact degree projection; the old development records are retained separately and are not a baseline performance comparison. No frozen evidence was rewritten.

Full-pilot blocker and next discriminating work

The three priced pilots remain open in the linked problem record. The existing AWS CLI was located and used; STS returned NoCredentials. There is no connected EKS or workflow-dispatch tool. The inspected main workflow accepts ecbench specifications and builds ecbench, not this standalone crate. It does not turn these new components into supported experiment adapters. No cloud launch, image or execution is claimed, and no identity, pool cap, frozen workload or original deadline was changed.

Next: integrate exact transfer and pre-elimination into the fully charged native query harness; build the algebraic quotient oracle; then run matched baseline/candidate/held-out queries with all costs, failures, lifting and verification under the authorized EKS contract. No smaller control here discharges the original ell8/ell9/m4 or quartic requests. Keep the original 29 October 20:02:21 UTC deadline and USD100 total; do not silently reset either.

Sources, novelty and graphs

The scope check read the 147-lead atlas and relevant hybrid-guess, cofactor quotient and isogeny-yield records. Its recorded review is 24 October and newer work remains unreconciled; this is not global novelty proof. Update those entries at the next scoped atlas review without changing their old closures. The contribution here is a checked application and executable certificate path for the three nominated experiments, not discovery of these standard algebraic identities or a new generic ECDLP algorithm.

- Existing source: examples/ic_algebraic_followup.rs, src/cryptanalysis/binary_semaev_s4.rs, examples/ic_projected_query_law.rs.
- Kosters--Yeo, https://arxiv.org/abs/1503.08001 section 4: first-fall degree can stay small while solving degree increases.
- Huang--Kosters--Yang--Yeo, https://arxiv.org/abs/1505.02532: zero-dimensional theorem hypotheses and explicit ECDLP limitation.
- Galbraith, chapter 25, https://www.cambridge.org/core/books/abs/mathematics-of-public-key-cryptography/isogenies-of-elliptic-curves/08E8C9D099ABC5FFBE15C6DD8469C985 . The earlier author-PDF access gap remains; no new full-PDF read is claimed.
- SageMath primary software documentation for Frobenius maps and finite-curve characteristic polynomials, read 29 October: https://doc.sagemath.org/html/en/reference/arithmetic_curves/sage/schemes/elliptic_curves/hom_frobenius.html and https://doc.sagemath.org/html/en/reference/arithmetic_curves/sage/schemes/elliptic_curves/ell_finite_field.html .

The algebraic/generic decomposition distinction in current main AGENTS is preserved. Truth-table/group enumeration here is a correctness reference, not algebraic IC progress. The current-head progress-timeline contract was checked: no cold IC/rho ratio or admitted cost exists to plot. The canonical scoreboard/leaderboard/progress figures and the historical closure-ratio figure are therefore unchanged. The new diagrams show component dimensions and verification scope only; no performance, curve-roster or ledger row is promoted.